Guide for circRNA Detection using CircTools in R

February 27, 2026

1 Directory Structure

The recommended folder structure is:

```
EasyCirc/
 circRNA/
     circtools/
         data/                  % input FASTQs, genome, rRNA, annotation
         samples.txt            % sample sheet
         R/                     % R scripts for pipeline
         circtools_detect/      % generated input files for detection
         output/                % circRNA detection results
         <SampleName>/          % sample-specific processing
             flexbar/          % adapter-trimmed FASTQs
             rrna/             % rRNA-filtered FASTQs
             star/             % STAR mapping outputs
                mate1/        % mate1-specific STAR output
                mate2/        % mate2-specific STAR output
             bam/               % converted BAM files (optional)
```

2 Create Conda Environment

```
conda create -n easycirc_env python=3.8 -y
conda activate easycirc_env
```

3 Install Required Tools

```
conda install -c bioconda flexbar bowtie2 star samtools -y
```

Verify installation paths:

```
which flexbar
which bowtie2
which STAR
which samtools
```

4 Install CircTools

```
git clone https://github.com/dieterich-lab/circtools.git
cd circtools
pip install .
```

Check that the `circtools` command is available:

```
circtools --help
```

5 Prepare Required Files

Place all input files under `EasyCirc/circRNA/circtools/data/`:

- Genome FASTA: `data/genome/hg38.fa`
- Annotation GTF: `data/genome/hg38.gtf`
- rRNA FASTA: `data/human`$_rRNA.fasta$
- `RepeatMasker GTF: data/repeats/RepeatMasker.hg38.fixed.gtf`
- `FASTQ files: data/fastq/Sample1_R1.fastq.gz, Sample1_R2.fastq.gz`

6 Sample Sheet

Create a tab-delimited sample sheet `samples.txt`:

```
SampleName      FileName1       FileName2
TMD8-8h-PQR     data/fastq/TMD8-8h-PQR_R1.fastq.gz      data/fastq/TMD8
    ↪ -8h-PQR_R2.fastq.gz
TMD8-8h-DMSO    data/fastq/TMD8-8h-DMSO_R1.fastq.gz     data/fastq/TMD8
    ↪ -8h-DMSO_R2.fastq.gz
```

7 Run the R Pipeline

7.1 Source the pipeline scripts:

```
source("R/pipeline_functions.R")
```

7.2 Execute CircTools preprocessing:

```
run_circtools(
  sample_file = "samples.txt",
  base_dir = "EasyCirc/circRNA/circtools",
  genome_fasta = "data/genome/hg38.fa",
  gtf_file = "data/genome/hg38.gtf",
  rRNA_fasta = "data/human_rRNA.fasta",
  threads = 10,
  flexbar_path = "/path/to/flexbar",
```

```
  bowtie2_path = "/path/to/bowtie2",
  star_exe = "/path/to/STAR"
)
```

This will perform:

1. Adapter trimming with Flexbar
2. rRNA removal with Bowtie2
3. STAR mapping (joint and mate-specific)
4. SAM to BAM conversion

The directory `circtools_detect/` will be generated for the next step.

8 Run CircTools Detect

```
cd EasyCirc/circRNA/circtools

circtools detect \
  @circtools_detect/samplesheet \
  -mt1 @circtools_detect/mate1 \
  -mt2 @circtools_detect/mate2 \
  -B @circtools_detect/bam_files.txt \
  -D \
  -R data/repeats/RepeatMasker.hg38.fixed.gtf \
  -an data/genome/hg38.gtf \
  -Pi \
  -F \
  -M \
  -Nr 2 1 \
  -fg \
  -G \
  -A data/genome/hg38.fa \
  -O output
```

Results will be saved in `EasyCirc/circRNA/circtools/output/`.

9 Read Results in R

```
source("R/output_reader.R")

result <- read_circtools_output(
  circ_output_dir = "EasyCirc/circRNA/circtools/output",
  samples = c("TMD8-8h-PQR", "TMD8-8h-DMSO"),
  chrMT = TRUE,
  save_csv = TRUE
)

head(result$circ_df)
head(result$circ_mtx)
```

Output: CSV files `circ_df.csv` and `circ_mtx.csv` in the output folder.

10 References

- CircTools GitHub: `https://github.com/dieterich-lab/circtools`
- CircTools Documentation: `https://docs.circ.tools/en/latest/Installation.html`